Dhanush M

Data Scientist · ML Engineer — LLM & Applied AI

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Built production LLM and computer-vision systems **from the ground up** — a multi-agent site-inspection platform and a multilingual voice-to-structured-data service — owning each end to end, from detection model and prompt architecture through API and production hardening.

EXPERIENCE

Slate Technologies — Construction Technology — Feb 2025 – Present

Assistant Software Engineer Jul 2026 – Present **Trainee Engineer** Jun 2025 – Jun 2026 **Intern** Feb 2025 – May 2025

Visual Intelligence — Automated Site Safety & Quality Inspection BUILT FROM SCRATCH

- **Built and iterated the hazard/defect detection model across three architectures** — SAHI-tiled YOLO → RF-DETR → TorchVision Faster R-CNN — adding tiled inference for high-resolution site imagery, S3-backed model loading and caching, and a resolution-independent blur gate that rejects unusable frames; consolidated safety and quality detection into a single inference pass served through both synchronous and batched asynchronous APIs.
- **Layered a VLM and LLM-agent stack over raw detections:** vision-language labeling that turns detections into semantic issue descriptions mapped to a deterministic CSI-code and quality taxonomy, plus a **six-stage LLM agent chain** (parse → sentence-split → rule extraction → filtering → mapping → deduplication) that converts each project's uploaded safety and quality documents into project-specific detection rules, replacing a fixed generic rule set.
- **Engineered the production video pipeline:** database-driven stage orchestration with per-stage status tracking, video compression, and ORB feature-similarity frame deduplication to avoid redundant inference on near-identical frames; authored the Dockerfiles and AWS Lambda trigger/runner scripts handed to DevOps for release.

STACK Python · PyTorch / TorchVision · RF-DETR · YOLO + SAHI · OpenCV · FastAPI · OpenAI API · PostgreSQL + pgvector · AWS S3, Lambda

Voice-to-Form-Fill — Multilingual LLM Speech-to-Structured-Data SOLE AUTHOR

- **Designed and built the full service** converting dictated site speech into submit-ready structured records: a **two-agent LLM pipeline** — intent matched against the tenant's live form types, then field values extracted against that form's real schema — preceded by an LLM normalization stage that handles **code-mixed input** (Tamil, Hindi, English and transliterated variants) while preserving construction terminology.
- **Cut end-to-end latency** by running the form-selection LLM call and the database schema prefetch concurrently on a thread pool, and added deterministic post-processing that resolves extracted names — assignees, dropdown options, site locations — to internal entity IDs, discarding unmatched values rather than emitting bad data.

STACK Python · Flask · OpenAI API (sync + async) · asyncio · ThreadPoolExecutor · PostgreSQL

Document Intelligence — Drawing & Specification Extraction

- **Built hierarchical LLM-agent extraction pipelines** for construction drawing sets (sheet classification, discipline, drawing-set and version extraction with bulk backfill) and technical specification documents (table-of-contents-aware parsing, dynamic sub-section extraction, PDF section splitting).
- **Authored the shared ingestion layer** — relational, graph, and vector database clients — that both parsers write into, feeding one downstream retrieval and search index.

Analytics Data Platform & Multi-Tenant APIs

- **Designed a multi-tenant analytics schema** (data views, dashboards, chart definitions with role-based permissions) plus denormalized reporting views, and **built the backend API serving it** — hardened with JWT-verified tenant/project scoping that rejects cross-tenant parameter mismatches and AES-256 encryption of stored queries.
- **Built an event-driven media pipeline in-database:** trigger functions converting site uploads into queued processing records and firing webhooks into the computer-vision service, with retry and timeout tuning for long-running inference.

AI Site Planning

- **Improved LLM-driven site layout generation** by replacing a fixed max-height heuristic with **dimension-based 3D model matching** — adding a model-dimension lookup and reworked classification prompt so generated building footprints resolve to models that physically fit the plot; also shipped an automated site flythrough with spline-interpolated camera pathing and geospatial coordinate projection.

SKILLS

Languages	Python · SQL
ML & CV	PyTorch · TorchVision · RF-DETR · YOLO / SAHI tiled inference · OpenCV · object detection · inference optimization
LLM & Applied AI	OpenAI API · vision-language (VLM) pipelines · multi-agent pipeline design · prompt engineering · structured extraction · taxonomy mapping
Data	PostgreSQL · pgvector · Neo4j · vector / graph / full-text retrieval · Hasura GraphQL · PostGIS · pandas
Platform	FastAPI · Flask · AWS S3 & Lambda · Docker · deployment scripting for DevOps handoff

EDUCATION

BE, Computer Science and Engineering — Sona College of Technology — **2021 – 2025**